Echo Grade Justification
Natesh Oad
November 16, 2024

Design and Documentation:

The Echo team has generated two diagram documents, these are an overall class structure diagram and a sequence diagram for sending messages from one user to another. Starting with the design of the class structure diagram, it is nicely formatted into sub sections, these include: Game and logic, User profile, Authentication and Networking, Leaderboard, and GUI. The diagram when taken together represents various aspects of the structure that Echo team strives for.

However, when creating a class structure with immense coverage the team lacks details in its class relationships. The diagram does not include proper relationships where appropriate, often utilizing dependencies rather than using other forms of aggregation such as composition. In other cases, the Echo team creates identical classes with different features and attributes that create confusion for readers and programmers. Classes should be formatted and named in a consistent manner to enable a project-wide usage, and if specific classes call for identical classes they should be programmed under their respective packages. The team often groups large abstract concepts into one particular class such as “Networking” which offers a broad level of services to other classes. This approach is highly flawed since coupling class together in this manner creates room for error and is highly suggested to instead split the class into subsections, for example “UserInteractionNetwork” for streamlined user-to-user communication. Echo teams can benefit from transferring code to their respective class for better code organization.

Likewise, to present multiple relationships with classes the team should look into multiplicity UML formats which can be used to simplify the diagram further. Additionally, the GUI components are too vague and inclusive making it difficult to create boundaries for programmers with no clear indication of class-to-class relationships. Echo team makes use of java standard and third party libraries that necessitate justification; libraries are clearly represented however their use cases have not been disclosed. Echo team should note that class structure diagrams are for reader and programmer readability, that give programers an preliminary starting point and is shared by multiple individuals and should be designed as so. The class structure diagram struggles with readability and often requires extensive back-tracking to figure out class descriptions and relationships, it is essential that the team include labeled arrows and move sections into their independent class structure diagram. These considerations need to be implemented mindfully since the class structure diagram is at the heart of code production and requires insights from various teams.

The sequence diagram for sending messages is designed with simplicity in mind and eliminates backend code for improved readability. The diagram should remove the parallel frame since the processes require an input therefore code execution does not parallel. For better readability purposes and for programmer convenience describing the specific code returns and arrow labels (and lifeline descriptions) are necessary. The colour background does allow for increased reliability, however

highlighting important elements of the diagram. Additionally, the team can implement actors with the diagram which enable programmers to correctly identify the use case, and design appropriately. The actor here is the end-user that sends a message to another user, which can be captured through an additional lifeline before the beginning of the dispatch. Along with an actor, team must closely follow the UML format, which includes underlining the labels to indicate an object for “Networkloop”. Otherwise, the team has done a great job in conveying as much details as possible in the most simple way possible, and may look into using such an approach in creating additional diagrams.

Planning and timeline: N/A

Justification: The letter grade of B+ shows that the team strives to show the mechanics of the projects in intermediate level of details. While the team has created an extensive class structure diagram it lacks in details that may lead developers to face future conflict which may affect certain proposed timelines. Team should look into resources when creating such documents to make sure that the design aligns with UML format and is organized into independent documents to improve readability. Sequence diagrams are however designed with simplicity and capture the essence in a readable manner, and suggested to create additional diagrams using similar approaches. With such flaws as described above in a document that is shared across iterations and individuals, it is necessary that the team take time to review documents to avoid similar issues in other areas. As such, a letter grade of B+ shows the team's efforts for creating an intuitive design, however one that requires rework to align with proposed expectations.